# MOBILE PLAN COST OPTIMIZATION

Python Programming Case Study

## 1. Problem Statement

Mobile users have different data limits, validity periods and plan prices. Selecting a suitable plan within a given budget can be difficult. This program compares available plans, calculates the cost including any extra data usage, and identifies the lowest-cost plan that meets the user's requirements.

## 2. Objective

To develop a Python program that accepts required data usage, validity days and budget, evaluates available mobile plans, checks plan eligibility and displays the most economical eligible plan.

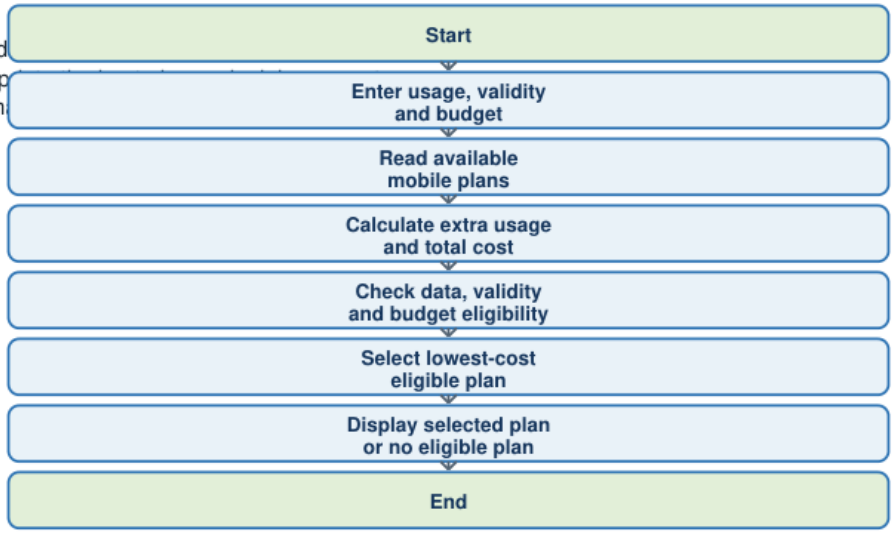
## 3. Assumptions

- Usage is entered in GB and validity in days.
- Each plan has a fixed data limit, validity, base price and extra-usage rate.
- Extra usage = max(0, required usage − plan data limit).
- Total cost = plan price + (extra usage × extra rate).
- A plan is eligible only if its data limit and validity meet the requirements and its total cost is within budget.
- If multiple plans are eligible, select the one with the lowest total cost.

## 4. Algorithm

1. Start and input required data usage, validity days and budget.
2. Store the available plans with their data limits, validity, prices and extra-usage rates.
3. Set best plan to None and minimum cost to infinity.
4. For each plan, calculate extra usage and total cost.
5. Check whether the plan meets the data, validity and
6. If eligible and cheaper than the current minimum, up
7. Display the best plan and minimum cost, or show th
8. Stop.

## 5. Flowchart

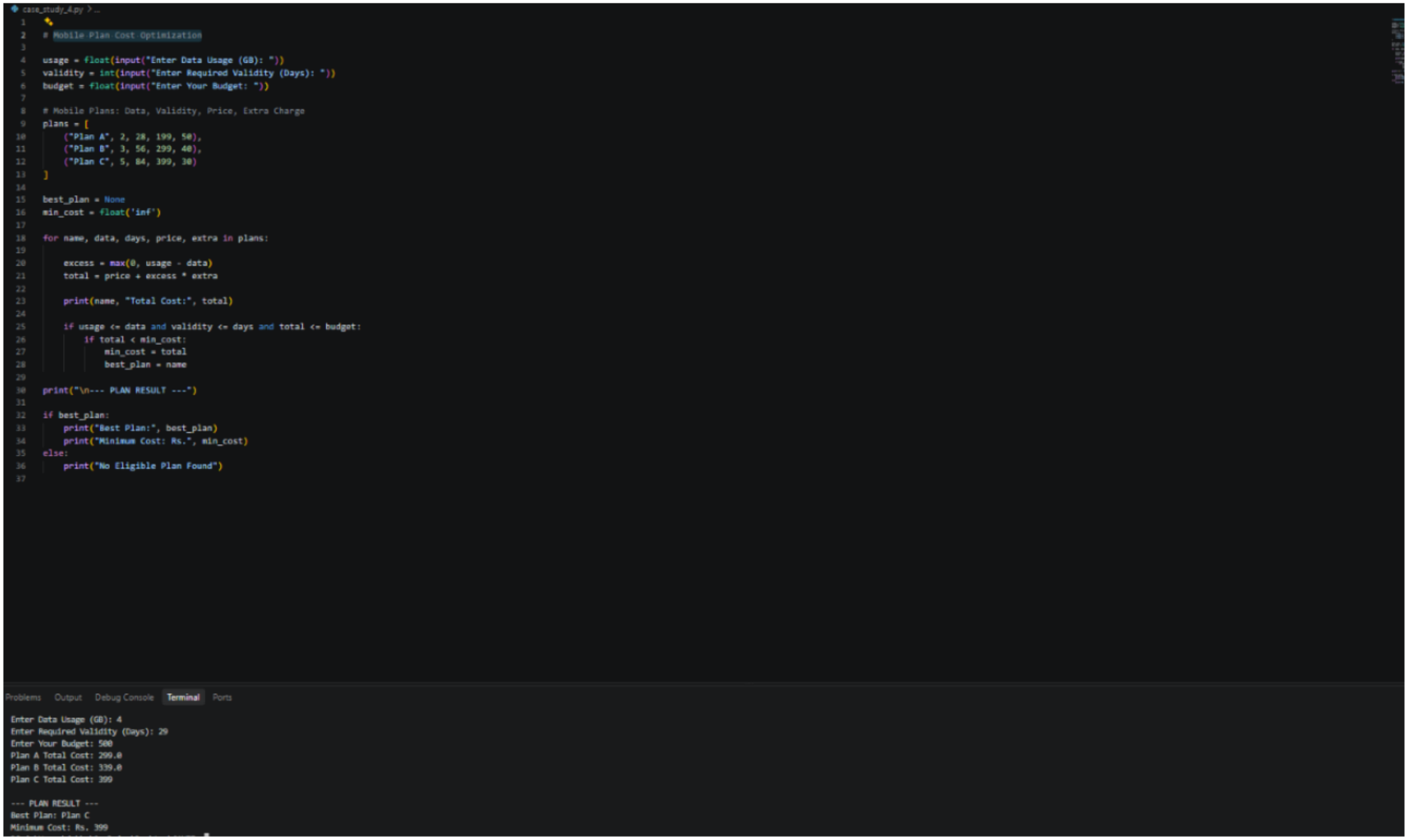



## 6. Pseudocode

**BEGIN**
INPUT usage, validity, budget
best_plan ← NONE; min_cost ← INFINITY
**FOR EACH** plan IN plans:
excess ← MAX(0, usage − plan.data)
total ← plan.price + excess × plan.extra_rate
**IF** usage ≤ plan.data AND validity ≤ plan.days AND total ≤ budget:
**IF** total < min_cost: min_cost ← total; best_plan ← plan.name
**IF** best_plan is not NONE: DISPLAY best_plan, min_cost
**ELSE:** DISPLAY "No Eligible Plan Found"
**END**

# 7. Python Source Code and Output

Original screenshot supplied for this case study



```python
# Mobile Plan Cost Optimization

usage = float(input("Enter Data Usage (GB): "))
validity = int(input("Enter Required Validity (Days): "))
budget = float(input("Enter Your Budget: "))

# Mobile Plans: Data, Validity, Price, Extra Charge
plans = [
    ("Plan A", 2, 28, 199, 50),
    ("Plan B", 3, 56, 299, 40),
    ("Plan C", 5, 84, 399, 30)
]

best_plan = None
min_cost = float('inf')

for name, data, days, price, extra in plans:

    excess = max(0, usage - data)
    total = price + excess * extra

    print(name, "Total Cost:", total)

    if usage <= data and validity <= days and total <= budget:
        if total < min_cost:
            min_cost = total
            best_plan = name

print("\n--- PLAN RESULT ---")

if best_plan:
    print("Best Plan:", best_plan)
    print("Minimum Cost: Rs.", min_cost)
else:
    print("No Eligible Plan Found")
```

```
Enter Data Usage (GB): 4
Enter Required Validity (Days): 29
Enter Your Budget: 500
Plan A Total Cost: 299.0
Plan B Total Cost: 339.0
Plan C Total Cost: 399

--- PLAN RESULT ---
Best Plan: Plan C
Minimum Cost: Rs. 399
```

## 8. Conclusion

The Mobile Plan Cost Optimization program demonstrates Python loops, conditional statements, lists and arithmetic operations. It compares plan costs against the user's data, validity and budget requirements, then selects the lowest-cost eligible plan. This helps users make a more informed and economical mobile plan choice.